

Name: \_\_\_\_\_



Exit quiz

## Securing understanding of equality

1 Which of these is the best definition for like terms? Tick **1** correct answer

- ☐ Terms which when added, sum to zero.
- ☐ Terms with the same coefficient for each variable.
- ☐ Terms with the same set of variables and corresponding exponents.
- ☐ Identical terms which have the same coefficient and same variables.
- ☐ Terms with the same variable and different exponents.

2 These 5 bar models represent operations performed on the equation  $x + 9 = 22$ . Match the bar models labelled A, B, C, D and E with the equivalent equation they represent. Write the correct letter in each box

A

|     |   |     |   |
|-----|---|-----|---|
| $x$ | 9 | $x$ | 9 |
| 22  |   | 22  |   |

B

|     |   |     |   |
|-----|---|-----|---|
| $x$ | 9 | $x$ | 9 |
| 22  |   | $x$ | 9 |

C

|     |   |     |   |     |   |
|-----|---|-----|---|-----|---|
| $x$ | 9 | $x$ | 9 | $x$ | 9 |
| 22  |   | $x$ | 9 | $x$ | 9 |

D

|     |   |     |   |     |   |
|-----|---|-----|---|-----|---|
| $x$ | 9 | $x$ | 9 | $x$ | 9 |
| 22  |   | 22  |   | 22  |   |

E

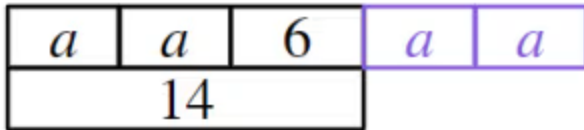
|     |   |    |
|-----|---|----|
| $x$ | 9 | 22 |
| 22  |   | 22 |

|   |   |
|---|---|
| a | A |
| b | B |
| c | C |
| d | D |
| e | E |

|  |                                  |
|--|----------------------------------|
|  | $x + 9 + x + 9 = 22 + x + 9$     |
|  | $x + 9 + 2x + 18 = 22 + 2x + 18$ |
|  | $2(x + 9) = 2(22)$               |
|  | $3(x + 9) = 3(22)$               |
|  | $x + 9 + 22 = 22 + 22$           |



- 3 This bar model represents the initial equation  $2a + 6 = 14$ . What needs to be done to the bottom bar to maintain equality? Tick **1** correct answer



- ☐  $\times 2$   
☐  $+ 14$   
☐  $+ 2a$   
☐  $+ 6$   
☐  $\times a$

- 4 If  $x + 5 = 2$ , which of these equations have maintained equality? Tick **3** correct answers

- ☐  $2x + 5 = 4$   
☐  $2x + 5 = x + 2$   
☐  $x + 3 = 0$   
☐  $x + 12 = 10$   
☐  $\frac{x + 5}{2} = 1$

- 5 If  $a = b$ , which of the following are true? Tick **2** correct answers

- ☐  $a - c = b - c$   
☐  $ac = \frac{b}{c}$   
☐  $a + c = b - c$   
☐  $ac = bc$   
☐  $a + c = b + d$



- 6 Which expression could fill the blank on the right hand side of this equation to maintain equality? Tick **3** correct answers

$$a + b = 5$$

$$a + b + a + b + a + b = \underline{\hspace{2cm}}$$

☐ 10

☐ 15

☐  $5 + a + b$

☐  $5 + a + b + a + b$

☐  $2a + 2b + 5$

